Netweb TECHNOLOGIES

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Empowering Compute, Network, Storage & AI

DDR5 RDIMM Memory Validation Report

Two-Campaign Test - 256 GB (2 TB) + 128 GB (1 TB)

Server / Host	172.16.13.217 (172.16.13.217)
Platform	Tyrone MDA200A2N-224 / 2x AMD EPYC 9135 (32C/64T)
Validation Tier	Qualification
Campaign Window	27 May 2026 (A 05:15-06:26, B 07:33-08:36 UTC)
Prepared For	Netweb Technologies India Limited
Date	2026-05-27

Overall Result: PASS

Executive Summary

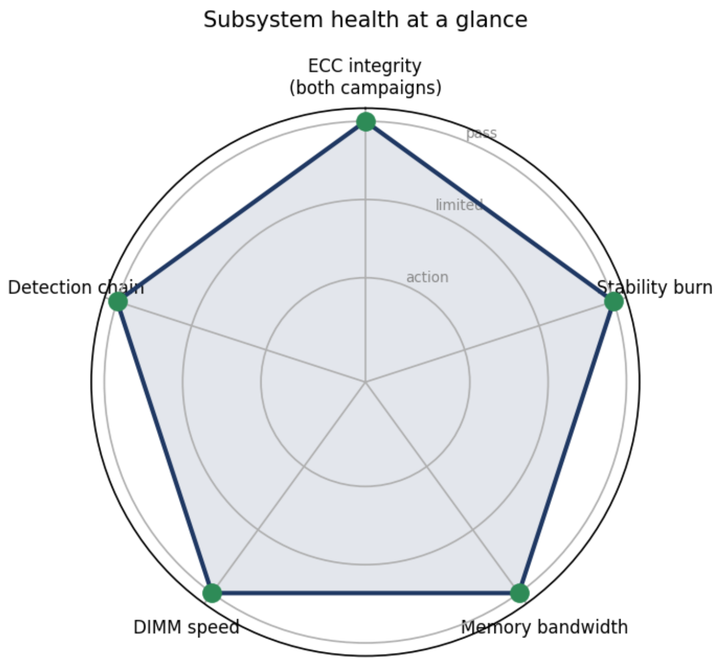
Two memory configurations were validated back-to-back on the same Tyrone MDA200A2N-224 server (172.16.13.217): Campaign A - 8x Samsung 256 GB DDR5-6400 RDIMMs (2 TB); Campaign B - 8x Samsung 128 GB DDR5-6400 RDIMMs (1 TB) after a physical swap. Each campaign ran a 30-minute stress-ng pattern-verify sweep followed by a 15-minute memtester burn that locked >=95% of RAM into physical memory.

All 16 modules across both campaigns are validated: zero ECC errors (0 CE / 0 UE across all 16 ranks), zero MCE events, and zero memtester failures across 90 minutes of combined testing.

Both SKUs are JEDEC-rated 6400 MT/s and both are platform-clamped to 6000 MT/s by the EPYC Turin 1DPC default. Aggregate STREAM bandwidth is essentially identical (~244 GB/s) in both campaigns - gated by the 4-of-12 channel population, not by module density. The capacity choice should be driven by working-set size, not bandwidth.

Validation Scorecard

Subsystem	Result	Detail	Verdict
ECC integrity (both campaigns)	0 CE / 0 UE across 16 ranks	0 MCE in 90 min; 16/16 ranks clean on mc0+mc1	PASS
Stability burn	>95% RAM locked, 0 failures	stress-ng verify (52 patterns) + 32x memtester	PASS
Memory bandwidth	~244 GB/s aggregate (both)	Per-channel ~30 GB/s - top of the 1DPC DDR5-6000 envelope	PASS
DIMM speed	6000 MT/s (stable)	Platform-clamped from rated 6400 (EPYC Turin 1DPC default)	PASS
Detection chain	BMC + BIOS + kernel agree: 8 DIMMs	Multi-bit ECC active; no slot/detection mismatch	PASS



Hardware Inventory

System

Vendor	Tyrone Systems
Model	MDA200A2N-224
Board	MH12XM

Hostname	172.16.13.217

CPU

Model	2x AMD EPYC 9135
Sockets	2
Cores Per Socket	16
Threads	64
Numa Nodes	2

Memory

Total	2 TB (A) / 1 TB (B)
Type	DDR5 RDIMM (Samsung)
Dimms Populated	8 of 24 (4-of-12 channels: P0/P1 A,C,G,I)
Configured Speed	6000 MT/s
Rated Speed	6400 MT/s
Speed Status	platform-clamped to 6000 (EPYC Turin 1DPC default)

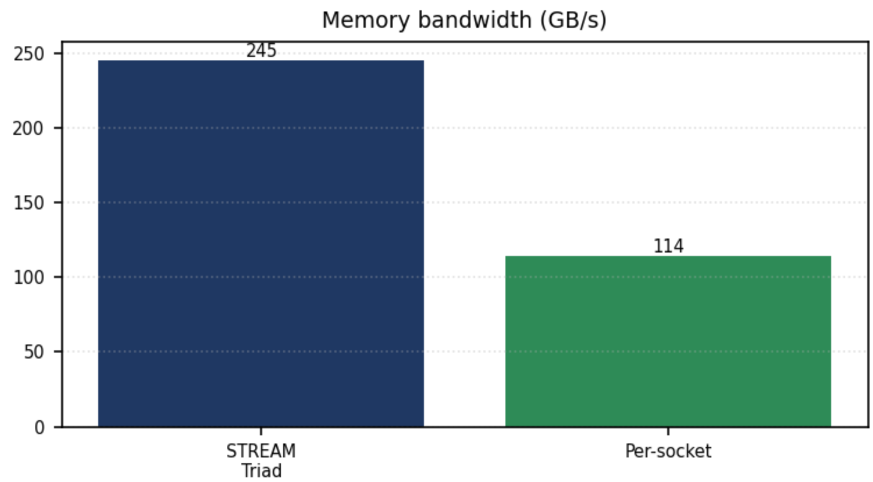
BIOS / Platform

Version	AMI ES312AMS.205T8 rev 5.35
Date	2026-03-26

Kernel / OS

Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-117-generic

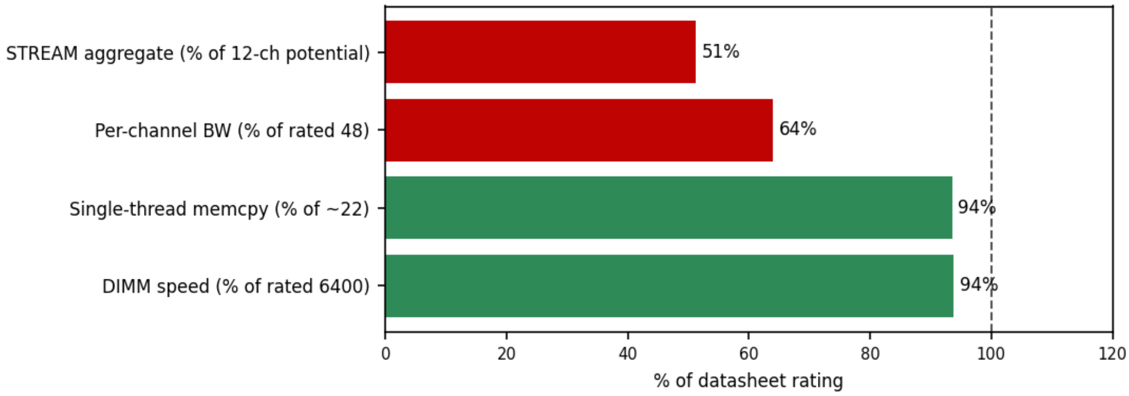
Memory



STREAM Triad (GB/s)	245.4
DIMM speed (MT/s)	6000 (rated 6400)

Specification Compliance

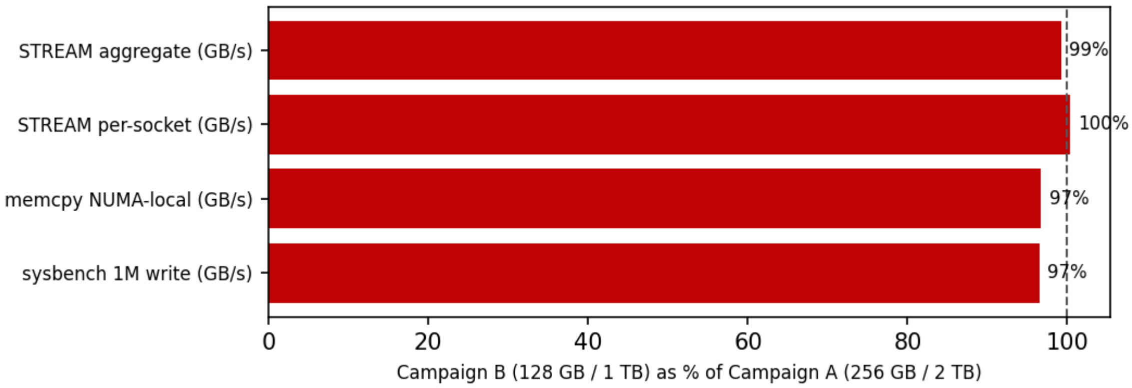
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
STREAM aggregate (% of 12-ch potential)	245.4	480	GB/s	51%
Per-channel BW (% of rated 48)	30.7	48	GB/s	64%
Single-thread memcpy (% of ~22)	20.57	22	GB/s	94%
DIMM speed (% of rated 6400)	6,000	6,400	MT/s	94%

Cross-Server Comparison

Same server, two DIMM populations swapped (256 GB vs 128 GB).



Metric	Campaign A (256 GB / 2 TB)	Campaign B (128 GB / 1 TB)	Delta
STREAM aggregate (GB/s)	245.4	243.6	-1%
STREAM per-socket (GB/s)	113.9	114.3	+0%
memcpy NUMA-local (GB/s)	20.57	19.89	-3%
sysbench 1M write (GB/s)	17.2	16.6	-3%

Aggregate bandwidth is identical (~244 GB/s) - gated by the 4-of-12 channel population, not module density. NUMA-remote penalty 1.69x (A) / 1.63x (B), within spec. Capacity is the only real differentiator between the two campaigns.

Component Health

BMC SEL (before → after): BMC SEL: 0 memory-related events during the 45-minute window in each campaign. IPMI sees exactly 8 populated DIMM thermal sensors (35-39 C); all voltage rails 'ok'.

Kernel log (dmesg) diff: 0 machine-check events across the 90-minute combined window. amd64_edac enumerated 16 ranks (8 UMCs x 2 CS); all CE=0 / UE=0 on mc0 + mc1 in both campaigns. No FAILURE/ERROR across 32 memtester logs per campaign.

Findings & Recommendations

Findings

- All 16 modules tested are electrically and structurally healthy (8x 256 GB + 8x 128 GB): 90 minutes of combined testing produced zero ECC errors at controller, rank and DIMM level.
- ECC is active in both configurations; amd64_edac enumerates 16 ranks per campaign, all clean; multi-bit ECC advertised in DMI Type 16. Kernel logged 0 machine-check events.
- mlock() succeeded on all 32 memtester workers in both campaigns - the system pins >=95% of RAM at either capacity (1.92 TB or 960 GB) without page-reclaim or OOM.
- Key insight: aggregate memory bandwidth is channel-population-limited (~244 GB/s at 4-of-12), not density-limited. Doubling capacity (A vs B) does not change bandwidth.
- All three management layers (BMC, BIOS, kernel) agree on 8 populated DIMMs - no detection discrepancies, no BMC-vs-BIOS mismatch.

Recommendations

- Populate more DIMM channels: today's 4-of-12 layout caps aggregate STREAM at ~244 GB/s; moving to 12-of-12 would scale toward ~500 GB/s with no module change - the highest-impact optimization regardless of module choice.
- Choose capacity by working set: use Campaign A (2 TB) for >=1 TB working sets (large in-memory DB, big model serving); use Campaign B (1 TB) when cost matters and the workload fits - bandwidth is identical.
- Run an 8-hour soak in the final configuration before shipping to production - raises the bar from infant-mortality clean to burn-in qualified.
- Install rasdaemon on the deployed system for continuous per-DIMM error history.

Conclusion

Area	Outcome	Verdict
8x 256 GB DDR5 (Campaign A, 2 TB)	Validated: 0 ECC, 0 MCE, >95% locked; ~245 GB/s aggregate	PASS
8x 128 GB DDR5 (Campaign B, 1 TB)	Validated: identical stability and bandwidth (~244 GB/s)	PASS
Bandwidth optimization	Populate 12-of-12 channels to scale toward ~500 GB/s	LIMITED

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Stress Ng	stress-ng --vm 32 --verify (52 patterns, 30 min)
Memtester	memtester 4.5.1 (32 workers, 15 min, 18-algo suite)
Bandwidth	stress-ng --stream; mbw; numactl variants
Latency	sysbench memory (4K, 16 threads, local + remote)
Edac	amd64_edac (mc0 + mc1, multi-bit ECC)
Bios	AMI ES312AMS.205T8 rev 5.35 (2026-03-26)
Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-117-generic

Campaign Window	27 May 2026 (A 05:15-06:26, B 07:33-08:36 UTC)
Note	Reconstructed from the validated source report into the CoreBench results.json contract.